

Was the Wealth of Nations Determined in 1000 BC?

Diego Comin, William Easterly, and Erick Gong (2010)

Presentation by Maciej Batkowski

July 3, 2026

Motivation & Research Question

- **Standard View:** Economic development researchers typically emphasize modern determinants: institutions, economic policies, human capital, and political factors. When history is discussed, it is usually limited to the Industrial Revolution or the effects of colonial regimes[cite: 26].
- **The Gap:** Could this discussion be missing a much more long-run dimension? Is it possible that history as old as 1500 AD or older also matters significantly for today's national economic development?
- **Research Approach:** Is technology persistent? To explore this, the authors assemble a new dataset on the history of technology over 2,500 years (1000 BC, 0 AD, 1500 AD) prior to extensive European colonization. The goal is to see if technological differences among predecessors to modern nations predict current per capita incomes and technology adoption.

Assembling the Dataset

- **Methodology:** The authors mapped ancient civilizations to modern national borders using the CIA World Factbook. If multiple cultures resided within a modern border, the culture with the *highest* technology level was used to represent that country.
- **1000 BC (113 countries) & 0 AD (135 countries):**
 - Derived from the *Atlas of Cultural Evolution* (ACE).
 - Evaluates 12 basic technologies across 5 sectors (e.g., tools, agriculture, communication).
 - **Data Assumption:** "Historic" civilizations (e.g., China, Rome) were assumed to have maximum technology scores by default, as the ACE only tracked prehistoric cultures.
- **1500 AD (113 countries):**
 - Coded entirely by the authors using over 200 historical sources.
 - Expands to 24 frontier technologies (e.g., firearms, magnetic compasses, printing).
 - *Crucial Note:* Evaluates pre-colonial technology, completely excluding technology transferred by Europeans after exploration began around 1500.

Methodology: The Extensive Margin

Due to the difficulty of sourcing old data, historical technology is measured strictly on the **extensive margin** (binary presence or absence), rather than intensity. Technologies tracked

across 5 sectors (they are equally weighted):

- **Military:** Bronze weapons, iron weapons, firearms, field artillery.
- **Agriculture:** Hand cultivation vs. plow cultivation.
- **Transportation:** Pack animals, wheels, magnetic compass, ocean-going ships.
- **Communications:** Writing, paper, movable block printing.
- **Industry:** Pottery, metalwork, steel.

Issue: what if a country had many cultures with varying technology level?

Coding technology presence

TABLE 2—CODING CONCORDANCE BETWEEN “ACE” AND THE TECHNOLOGY ADOPTION DATASET

“ACE” dataset	Technology dataset for 1000 BC and 0 AD*
Writing and records	Communication
1 = None	
2 = Mnemonic or nonwritten records	0, 1
3 = True writing	0, 1
Technological specialization	Industry
1 = None	
2 = Pottery	0, 1
3 = Metalwork (alloys, forging, casting)	0, 1
Land transport	Transportation
1 = Human only	
2 = Pack or draft animals	0, 1
3 = Vehicles	0, 1
Agriculture	Agriculture
1 = None	0
2 = 10 percent or more, but secondary	1
3 = Primary	2
Military	Military
1 = Stone tools	
2 = Bronze tools	Bronze weapons: 0, 1
3 = Iron tools	Iron weapons: 0, 1

To test persistence into the modern era, the authors calculate a **Current Technology Index**. Unlike historical periods, modern technology diffuses globally very quickly, making the extensive margin useless. Instead, they measure the **intensive margin**.

- Tracks 10 technologies (Electricity, Internet, PCs, Aviation, etc.).
- Calculates the **Time Lag**: How many years ago the US had the target country's current adoption intensity.
- Normalized by the age of the technology.

Is technology persistent?

TABLE 7A—TECHNOLOGY IN 1500 AD AS FUNCTION OF TECHNOLOGY IN 0 AD OR 1000 BC

Dependent variable	Overall technology adoption level in 1500 AD									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Overall technology adoption level in 0 AD	0.782*** (8.36)		0.356*** (5.75)		0.614*** (7.14)		0.635*** (7.58)		0.691*** (6.15)	
Overall technology adoption level in 1000 BC		0.784*** (5.55)		0.242*** (4.92)		0.582*** (3.50)		0.584*** (3.43)		0.637*** (3.25)
Europe dummy			0.520*** (11.07)	0.689*** (17.08)						
Africa dummy			0.00481 (0.10)	0.193*** (4.41)						
Asia dummy			0.368*** (5.11)	0.537*** (9.11)						
America dummy			-0.00448 (-0.34)	0.0681* (1.90)						
Distance to equator					0.348 (0.57)	-0.0973 (-0.27)	0.424 (0.82)	-0.118 (-0.30)		
Distance to equator squared					0.590 (0.82)	1.280*** (2.75)	0.438 (0.72)	1.310** (2.71)		
Landlocked							-0.0979*** (-2.79)	0.0155 (0.38)	-0.101** (-2.50)	0.0104 (0.22)
Tropical dummy									-0.249*** (-3.09)	-0.215** (-2.26)
Constant	-0.0624 (-0.90)	0.128* (1.70)	0.0296*** (2.85)	0.0182 (0.56)	-0.129 (-1.64)	0.0881* (1.71)	-0.129* (-1.79)	0.0860 (1.62)	0.146 (1.46)	0.299** (2.13)
Observations	116	104	116	104	108	97	108	97	116	104
R ²	0.47	0.47	0.86	0.87	0.67	0.63	0.69	0.63	0.64	0.57

Ancient technology and modern technology and income

TABLE 8A—ANCIENT TECHNOLOGY MEASURES AND INCOME AND TECHNOLOGY TODAY

Dependent variables	Log income per capita in 2002			Current technology		
	(1)	(2)	(3)	(4)	(5)	(6)
Overall technology adoption level in 1000 BC	0.747* (1.87)			0.0851 (1.37)		
Overall technology adoption level in 0 AD		0.0895 (0.19)			0.0138 (0.14)	
Overall technology adoption level in 1500 AD			1.566*** (3.24)			0.221** (2.58)
Constant	8.196*** (28.22)	8.452*** (19.62)	7.786*** (22.72)	-0.590*** (-13.60)	-0.557*** (-6.89)	-0.655*** (-12.22)
Observations	104	123	111	109	130	115
R^2	0.03	0.00	0.18	0.02	0.00	0.12

Outliers

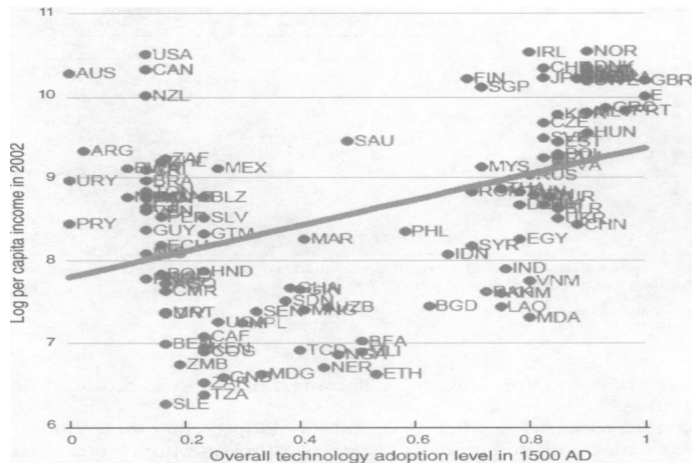


FIGURE 1. TECHNOLOGY IN 1500 AD AND CURRENT DEVELOPMENT

Addressing Outliers: The Shift to "Peoples"

The "Neo-Europe" outliers force a critical realization: **Technology does not necessarily persist in places; it persists in people.**

- **The Context:** Post-1500 AD saw unprecedented global migration, colonization, and population replacement.
- **The Solution:** Using the **Putterman & Weil (2008)** matrix, the authors calculate a weighted average based on where the *ancestors* of today's population lived in 1500 AD.

The Migration Adjustment Calculation:

$$T_i^{adj} = \sum_{j=1}^N (m_{i,j} \times T_j)$$

- T_i^{adj} : The adjusted 1500 AD technology score for modern country i .
- $m_{i,j}$: The fraction of country i 's current population whose ancestors lived in territory j in 1500 AD.
- T_j : The original 1500 AD technology score for territory j .

Ancient technology and modern technology and income

TABLE 8B—MIGRATION-ADJUSTED TECHNOLOGY MEASURES AND INCOME AND TECHNOLOGY TODAY

Dependent variables	Log income per capita in 2002			Current technology		
	(1)	(2)	(3)	(4)	(5)	(6)
Migration-adjusted technology level in 1000 BC	1.599*** (3.45)			0.211*** (3.3)		
Migration-adjusted technology level in 0 AD		2.303** (2.35)			0.418** (2.56)	
Migration-adjusted technology level in 1500 AD			3.261*** (6.76)			0.514*** (6.87)
Constant	7.697*** (23.46)	6.602*** (8.65)	6.544*** (18.6)	−0.662*** (−16.65)	−0.895*** (−7.85)	−0.862*** (−21.06)
Observations	104	123	111	109	130	115
R^2	0.12	0.11	0.5	0.07	0.12	0.4

Interpreting the Magnitude: Model Variations

The real-world magnitude of the 1500 AD effect depends heavily on the model specification, though it remains substantial across the board.

Model Specification	Coefficient (β)	Income Multiplier (e^β)
Place-Based (No Controls)	1.566	$\sim 4.8\times$ (Factor of 5)
Migration-Adjusted (No Controls)	3.261	$26.1\times$
Migration-Adjusted (+ Geo Controls)	1.770 to 2.590	$5.9\times$ to $13.3\times$

Summary of Results:

- **The Peak:** When adjusted purely for migration, shifting from 0 (no tech) to 1 (full frontier tech) multiplies modern income by over 26 times.
- **The Conservative Estimate:** When controlling for strict geographic factors (latitude, landlocked status, tropical climate), the effect dampens but remains highly significant.
- **Conclusion:** Even in the most rigorous models, ancestral technology multiplies modern income by a minimum factor of roughly 6.

Theoretical Framework: Why Does Technology Persist?

In Part III, the authors propose a framework to model technology adoption dynamics:

$$A_{ct} - A_{ct-1} = A_{ct-1}^{\gamma} e^{\theta_{ct}} + \epsilon_{ct}$$

- A_{ct} : Level of technology in country c at time t .
- γ : The strength of complementarity between previous technology and new technology.
- θ_{ct} : Return to adopting new technology (driven by other factors like institutions, geography, or population).

The core question revolves around γ : If $\gamma < 1$, initial advantages eventually vanish (convergence). Because the data shows long-run persistence, γ must be ≥ 1 .

The Alternative Hypothesis: Is it just an omitted variable?

An alternative theory argues that γ (direct tech persistence) is actually close to zero, and the persistence we see is just because θ_{ct} (the return to adoption) is persistently higher in some places.

The Population Test:

- Semi-endogenous growth models suggest that larger populations create more inventors, persistently driving up θ_{ct} .
- **The Test:** The authors run their regressions again, explicitly controlling for historical population size in 1500 AD.
- **The Result:** Historical technology remains a highly significant predictor of modern development, while historical population does not. Population scale alone cannot explain technological persistence.

The Ultimate Proof: Within-Sector Variation

How do we rule out other unmeasured historical variables, like 1500 AD institutions or cultural genetics, driving θ_{ct} ?

- **The Logic:** If a persistent factor like "good institutions" drove technology adoption, it would benefit all sectors (military, agriculture, etc.) symmetrically within a country.
- **The Identification Strategy:** The authors look at the variation of technology *within specific sectors*, stripping away the country-wide average adoption level (which controls for all nation-wide institutional or geographic advantages).
- **The Result:** Persistence holds **within individual sectors**. This proves that past technology directly lowers the cost of future technology in that same sector, confirming γ is a primary driver.

- **Deep Roots:** Technology as old as 1500 AD is a highly robust historical correlate of modern development, explaining up to 50% of the variance in global wealth today.
- **People, Not Places:** Technological persistence is bound to populations and their migration patterns, not the geographical soil.
- **The Cost of Adoption:** The evidence strongly supports a model where the cost of adopting modern frontier technologies falls significantly based on the existing stock of historical technology.